

MAN00035C

Quantitative Analysis in Business Studies

2026 – Week 4



UNIVERSITY
of York

School for Business
and Society



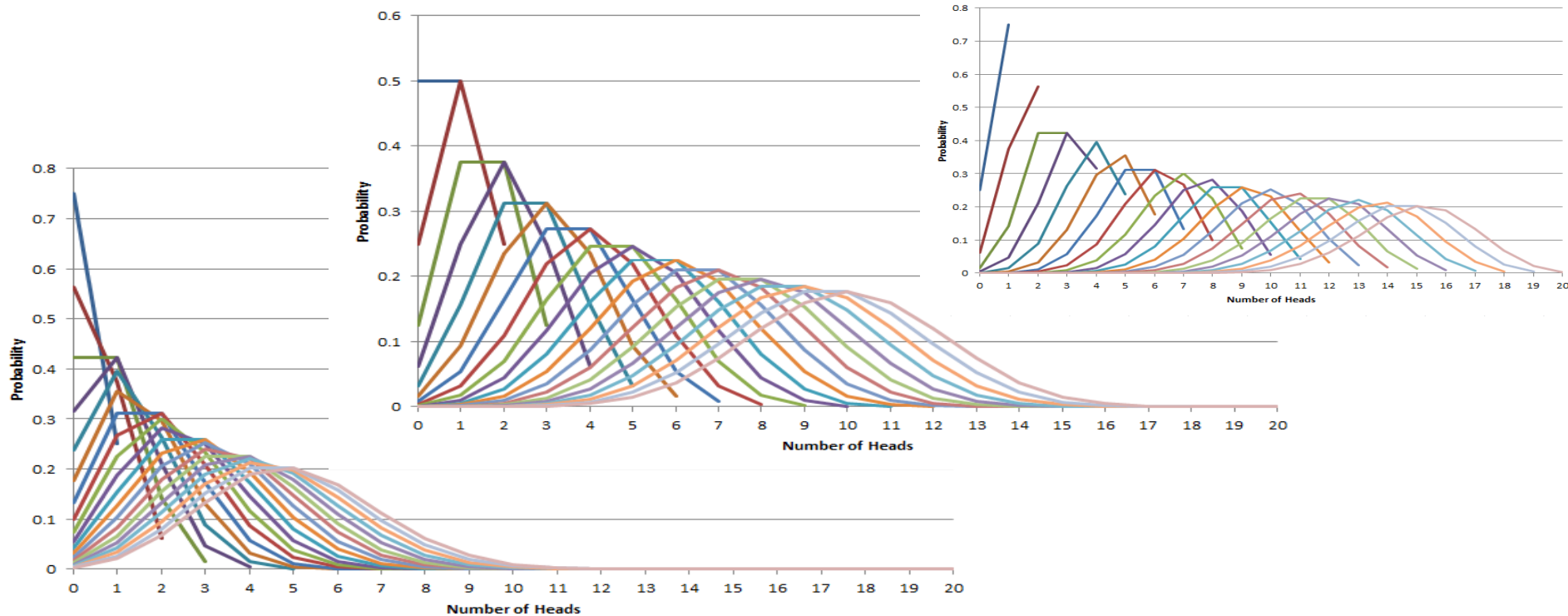
Data Sampling and Estimation

Week 4

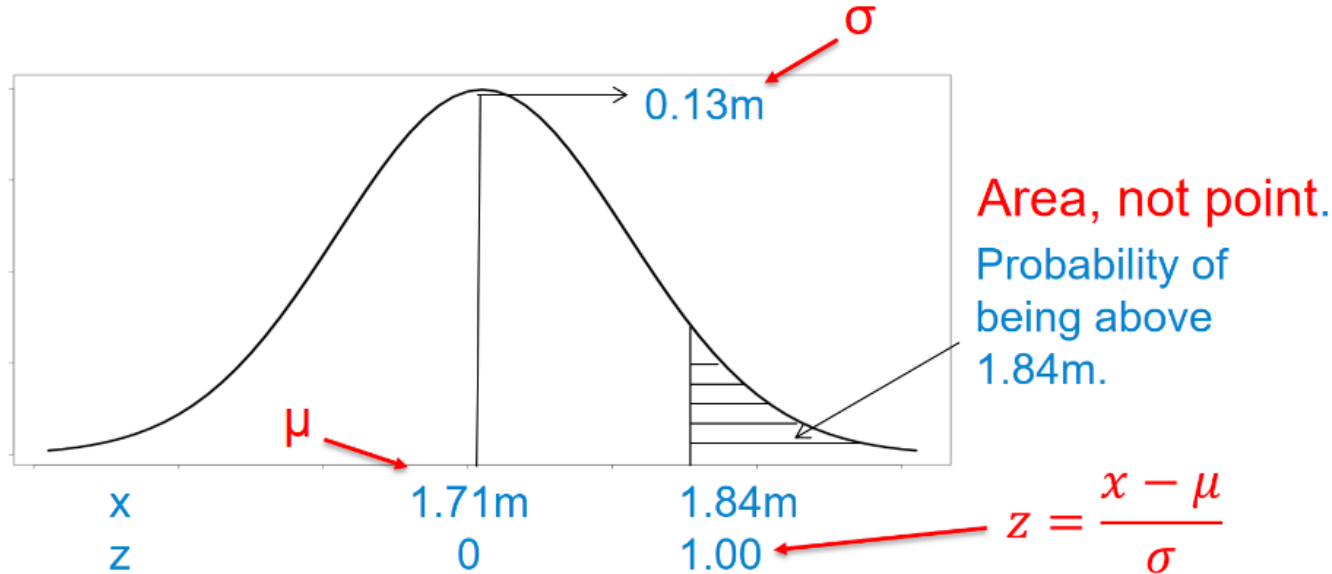
Last topic

- We covered the probability distributions:
 - Bernoulli (p)
 - Binomial (n, p)
 - Normal (μ, σ^2)
- We looked at the **mean**, μ , and **spread**, σ^2 , of each.
- We saw that the Binomial is the **sum** of n Bernoulli trials, and a Binomial **approximates the normal** distribution when n gets large.

Last topic



The normal distribution



A diagram of the normal distribution with the mean (1.71m) and standard deviation (0.13m) marked explicitly.

We're looking for the area above 1.84m, which corresponds to $z=1.00$.

This lecture

We're going to look at:

- Sampling
- Sampling distributions; The Central Limit Theorem
- Point estimation
- Confidence intervals

Sampling

Population and samples

- **Population:** a **collection of entities** (people, objects, events), or **collection values** (observations) of some variable of interest characteristic of the entities
- **Example 1:** We are interested in studying the height of students at the University of York. The population is the collection of **all students** from the UoY, or the collection of **'heights' of all** students at the UoY
- **Example 2:** We are interested in the performance of firm shares listed in all European stock exchanges. **All the firms**, or the **'prices' of all firm shares** listed in European stock exchanges is the population

Sampling

The full population might not be possible to observe

- Too time consuming, or too expensive,
- Or even not available, etc...

When we can't have the values for the full population we have to use a sample

Sample: subset (part) of the population, or subset of values of the variable of interest characteristic of the entities

- **Example 1:** the subset of UoY students at Alcuin college, or the collection of the Alcuin college students 'heights' is a sample from the UoY students

Sampling

- A sample should be **representative** of the population
- If not, we cannot make **inference** about the all population from the sample
- **Example 1:** Population - UoY students.
 - Characteristic/variable of interest – students' height
 - Sample: students who play in the UoY basketball team - **NOT representative!** Why?
- **Example 2:** Population – all firms listed in European stock exchanges
 - Characteristic/variable of interest – firms' share price
 - Sample: randomly selected subset of listed firms - **Representative?** Why?

Description vs Inference

Example: all 159 students enrolled in a given module:

	A	B	C	D	E	F	G	Formula bar
1	Gender ▾	Height (m) ▾	Weight (kg) ▾	Height Guess ▾	Weight Guess ▾	Breakfast? ▾	College ▾	Confident With Numbers ▾
2	Female	1.35	23	1.74	59	Yes	Goodricke	Agree
3	Female	1.45	34	1.83	80	Yes	Constantine	Disagree
4	Female	1.47	36	1.7	56	Yes	Constantine	Disagree

- We had the dataset for the all population and described the shape of the distribution of a characteristic of interest, e.g. height: **mean, spread, and graphs of distribution**
- If we had only a subset or **sample** from the students in the module, we could still **infer** from the sample about the height of all students in the module population

Statistical inference: is the procedure to reach a conclusion about a population characteristic based on the information contained in a sample from that population

What makes a sample good?

- A “good” sample needs to be “**representative**”.
- It needs to encode **everything relevant** to the characteristic of interest about the population
- What is relevant depends on the **purpose** of your investigation.
- **Example:** Interested in self-employed workers income:
 - **No data** on all self-employed workers => use a sample
 - Obtain data on income among self-employed workers across different:
 - Sectors, professions, regions, size of the business, etc

Primary vs Secondary

Primary data is any data you collect yourself for the intended purpose

- Example: You design and conduct interviews, run lab experiments, or send a survey
 - Control over quality and design (questions asked)
 - Expensive, time consuming, takes a lot of work

Secondary data is any data someone else has already collected

- Example: Stock prices available from *Yahoo Finance*
 - Lower cost, quick to obtain
 - You need to trust the provider, it might be biased or outdated

Different types of sampling

Random sampling: sample of elements drawn from a population such that every unit of the population has a **non-zero chance** of being selected

- The methods we'll look at are:
 - Simple random sampling
 - Systematic
 - Stratified
 - Cluster
- **Sampling unit:** unit on which the sampling is based. It can be
 - An individual element of the population: e.g. a firm, an employee, a student
 - A group of elements: e.g. household
- A **sampling frame** helps, this is a **list of all sampling units** of the population
 - It's not always essential, and may not be possible to have

Simple random sampling

Simple random sampling: out of everyone (everything) in the population, each element has an equal chance to be picked for inclusion in the sample

- We need a **sampling procedure plan** (who, what, where, when, how...)

- **Example:** if you want a sample of 10 manufactured mugs, and 1,000 come out of the kiln
 - Give each mug a number
 - Ask a computer to choose 10 random numbers between 1 and 1,000
 - Pick up the mugs with the corresponding numbers
 - Each mug has a (non-zero) 1% chance of being selected

Systematic sampling

- Convenient when population elements are e.g. **time stamped** or **happen sequentially**
- Assume we want a sample that is $1/M$ of the total population of size N
- We randomly select one of the first M items, and every M th item thereafter until we cover the entire population.

Example: We want a sample of 1 out of every 50 items out of a manufacturing process. Randomly select 1 out of the first 50, say number 3, and every 50th item after that (53, 103, 153, 203,...). After 1,000 items produced the sample will have $n=N/M=1,000/50=20$ items.

Stratified sampling

- Divide the population up into sub-groups (“**strata**”) that share a common characteristic – groups are **homogeneous**
- The characteristic should be something you’re interested in.
- Then use simple random sampling within each stratum.

Example: you want to see how **income** effects people's view of education, your strata should probably be based around their **income** (*i.e. a low-income group, a middle-income group, a high-income group*) or **profession**

Example: If you want to see how different **genders** view safety at night, you probably want to strata across **gender**

Stratified sampling – Example

- Examples may be helpful.

- Imagine we want to sample 50 students and get a good breakdown of their colleges and / or genders.

- We can do this in three ways:
 - by gender;
 - by college;
 - by both simultaneously.

Stratified sampling – Example

More precisely, then, it would be:

$$50 \times \frac{64}{159} = 20.13$$

and

$$50 \times \frac{95}{159} = 29.87.$$

and round to whole units (people).

		Gender			
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Male	64	40.3	40.3	40.3
	Female	95	59.7	59.7	100.0
	Total	159	100.0	100.0	

Stratified sampling – Example

By college, then:

- Consider trying to sample the person from Alcuin, although that would make them identifiable.
- Consider reducing Goodricke to 12.

College	Number	Percentage	Want to sample	Rounded
Alcuin	1	0.6%	$50 \times 0.006 = 0.3$	0
Constantine	38	23.9%	$50 \times 0.239 = 11.9$	12
Derwent	12	7.5%	$50 \times 0.075 = 3.8$	4
Goodricke	41	25.8%	$50 \times 0.258 = 12.9$	13
Halifax	13	8.2%	$50 \times 0.082 = 4.1$	4
James	25	15.7%	$50 \times 0.157 = 7.9$	8
Langwith	26	16.4%	$50 \times 0.164 = 8.2$	8
Vanbrugh	3	1.9%	$50 \times 0.019 = 0.9$	1
Total	159	100.0%	50	50

Stratified sampling - Example

By college AND gender at the same time:
The sample gave us:

	Men	Women	Total
Alcuin	1	0	1
Constantine	10	28	38
Derwent	7	5	12
Goodricke	17	24	41
Halifax	7	6	13
James	14	11	25
Langwith	7	19	26
Vanbrugh	1	2	3
Total	64	95	159

	Men	Women	Total
Alcuin	0.6%	0%	0.6%
Constantine	6.3%	17.6%	23.9%
Derwent	4.4%	3.1%	7.5%
Goodricke	10.7%	15.1%	25.8%
Halifax	4.4%	3.8%	8.2%
James	8.8%	6.9%	15.7%
Langwith	4.4%	11.9%	16.4%
Vanbrugh	0.6%	1.3%	1.9%
Total	40.3%	59.7%	100.0%

Stratified sampling - Example

For each cell, we divide by the population size (159), multiply by the sample size (50) and round:

	Men	Women	Total
Alcuin	0.3	0.0	0.3
Constantine	3.1	8.8	11.9
Derwent	2.2	1.6	3.8
Goodricke	5.3	7.5	12.9
Halifax	2.2	1.9	4.1
James	4.4	3.5	7.9
Langwith	2.2	6.0	8.2
Vanbrugh	0.3	0.6	0.9
Total	20.1	29.9	50

	Men	Women	Total
Alcuin	0	0	0
Constantine	3	9	12
Derwent	2	2	4
Goodricke	5	8	13
Halifax	2	2	4
James	4	3	8
Langwith	2	6	8
Vanbrugh	0	1	1
Total	20	30	50

Stratified sampling - Example

	Men	Women	Total
Alcuin	0.3	0	0.3
Constantine	3.1	8.8	11.9
Derwent	2.2	1.6	3.8
Goodricke	5.3	7.5	12.9
Halifax	2.2	1.9	4.1
James	4.4	3.459	7.9
Langwith	2.2	6.0	8.2
Vanbrugh	0.3	0.6	0.9
Total	20.1	29.9	50

	Men	Women	Total
Alcuin	0	0	0
Constantine	3	9	12
Derwent	2	2	4
Goodricke	5	8	13
Halifax	2	2	4
James	4	3	8
Langwith	2	6	8
Vanbrugh	0	1	1
Total	18	31	50

But there are problems... We have rounded down for EVERY male category. There isn't a good systematic way to fix this.

Cluster sampling

In **cluster sampling**, we select a sample of groups from the sampling frame and collect information on all the individual elementary units in the groups selected. Each group is called a **cluster**

- Divide the population into "natural" groups called clusters (like neighbourhoods, schools, or city blocks). Then, randomly choose a few entire groups and survey everyone inside them
- Efficiency: saves time and money
- The groups are **Heterogeneous** (each group should be a "mini-version" of the whole population)
- **Example:** We want to survey Tesco stores employees. It's too expensive to collect data from all stores. Instead, randomly pick 10 specific stores and interview every single person who works at those 10 locations

Interviews: Leading Questions

- Just like we had to think about how to choose a sample to keep it representative, we have to think about how to word a question to avoid bias.
- One issue is that of leading questions.
- <https://www.youtube.com/watch?v=G0ZZJXw4MTA&t=31sa>
- This clip from “Yes, Prime Minister” illustrates this nicely.





Sampling distributions

Population parameters vs sample statistics



- We say that a population has **parameters**, e.g.
 - μ is the mean
 - σ is the standard deviation
- Usually, we don't know the values of the population parameters

Example: For the population of all 159 students enrolled in a given module, we calculated the mean, variance and standard deviation of the students' height:

- $\mu = 1.71m$
- $\sigma^2 = 0.01612091$
- $\sigma = 0.12696813m$

Population parameters vs sample statistics



- Suppose that we only had a random sample of size n from the population of all $N=159$ students ($n < N$)
- We don't have the all dataset $\Rightarrow$ We cannot calculate μ or σ
- But we can use the sample to make **inference** about the larger population
- And we can calculate the mean, standard deviation, and other quantities of interest from the sample
- We call these quantities calculated from the sample observations **sample statistics**
- We use different notation to differentiate between population parameters and sample statistics

Population parameters vs sample statistics



Different notation to distinguish between (all) population parameters and (smaller) sample statistics:

	Population parameters	Sample statistics
Mean	μ	$\bar{x}$
Standard deviation	σ	s
Proportion	π	p

Example

- I want to learn about the **height** of the students enrolled in the module in the previous example.
- Suppose that I have the **list** of all students, but **not** their height
- I use simple random sampling to select a **sample** of say $n=20$ students
- I ask those students to tell me (anonymously) their height
- I obtain the sample S:

Sample, S = (1.69,1.7,1.72,1.72,1.73,1.75,1.76,1.8,1.81,1.83,1.83,1.84,1.89,1.9,1.93,1.54,1.58,1.64,1.69,1.73)

- I want to **estimate** the mean (μ) and the standard deviation (σ) of the height of all students **using the sample statistics**, sample mean ($\bar{x}$) and sample standard deviation (s), calculated from the sample S

Sampling distribution of the sample mean

- Question: is the sample mean ($\bar{x}$) a **good guess** or **good estimator** of the population mean (μ)?
- Problem:
 - There are many different possible random samples of size n from the population of size N . Remember that typically $n \ll N$
 - Potentially, all possible random samples have a different sample mean
- Hence,
 - The sample mean ($\bar{x}$) is a **random variable**, $\bar{X}$
 - The sample mean has its own probability distribution: the **sampling distribution of the sample mean**

Sampling distribution of the sample mean

- We could consider all possible random samples of size n from a population with mean μ and variance σ^2
- Calculate the sample mean, variance and standard deviation of all samples

We would observe that:

- The mean of the calculated sample means is exactly the mean of the population

$$E(\bar{X}) = \mu$$

- The variance of the calculated sample means is NOT exactly the variance of the population

$$\text{var}(\bar{X}) = \sigma^2 / n$$

Sampling distribution of the sample mean

- We could consider all possible random samples of size n from a population with mean μ and variance σ^2
- Calculate the sample mean, variance and standard deviation of all samples

We would observe that:

- The standard deviation of the calculated sample means is NOT exactly the standard deviation of the population

$$\sqrt{\text{var}(\bar{X})} = \sigma / \sqrt{n}$$

- $\sigma / \sqrt{n}$ is called the **standard error of the sample mean** or SEM

- If the population has a normal distribution, then the **sampling distribution of the sample mean is normal**

$$\bar{X} \sim N(\mu, \sigma^2 / n)$$

Sampling distribution of the sample mean

Example: Consider the population of the 159 students enrolled on the given module

- We are interest in the ‘height’ of the students
- From the random sample ($n=20$)
(1.69,1.7,1.72,1.72,1.73,1.75,1.76,1.8,1.81,1.83,1.83,1.84,1.89,1.9,1.93,1.54,1.58,1.64,1.69,1.73)
- We calculate the sample mean $\bar{x}$, and the standard error of $\bar{x}$:
 - $\bar{x} = \sum_{i=1}^{20} x_i / 20 = 1.754$, and
 - $SEM = \sigma / \sqrt{n} = 0.12696813 / \sqrt{20} = 0.00634841$

Sampling distribution of the sample mean

- If the population has a normal distribution, then the sampling distribution of the sample mean is normal

$$\bar{X} \sim N(\mu, \sigma^2 / n)$$

- What if the population is not normally distributed?
- Then we have to use the **Central Limit Theorem** to know the sampling distribution of the sample mean

The Central Limit Theorem

- Given a population with mean μ and variance σ^2

if the sample size n is sufficiently large

then the sampling distribution of the sample mean **approximates a normal distribution** with mean μ and variance σ^2 / n

$$N(\mu, \sigma^2 / n)$$

no matter what the actual population distribution is

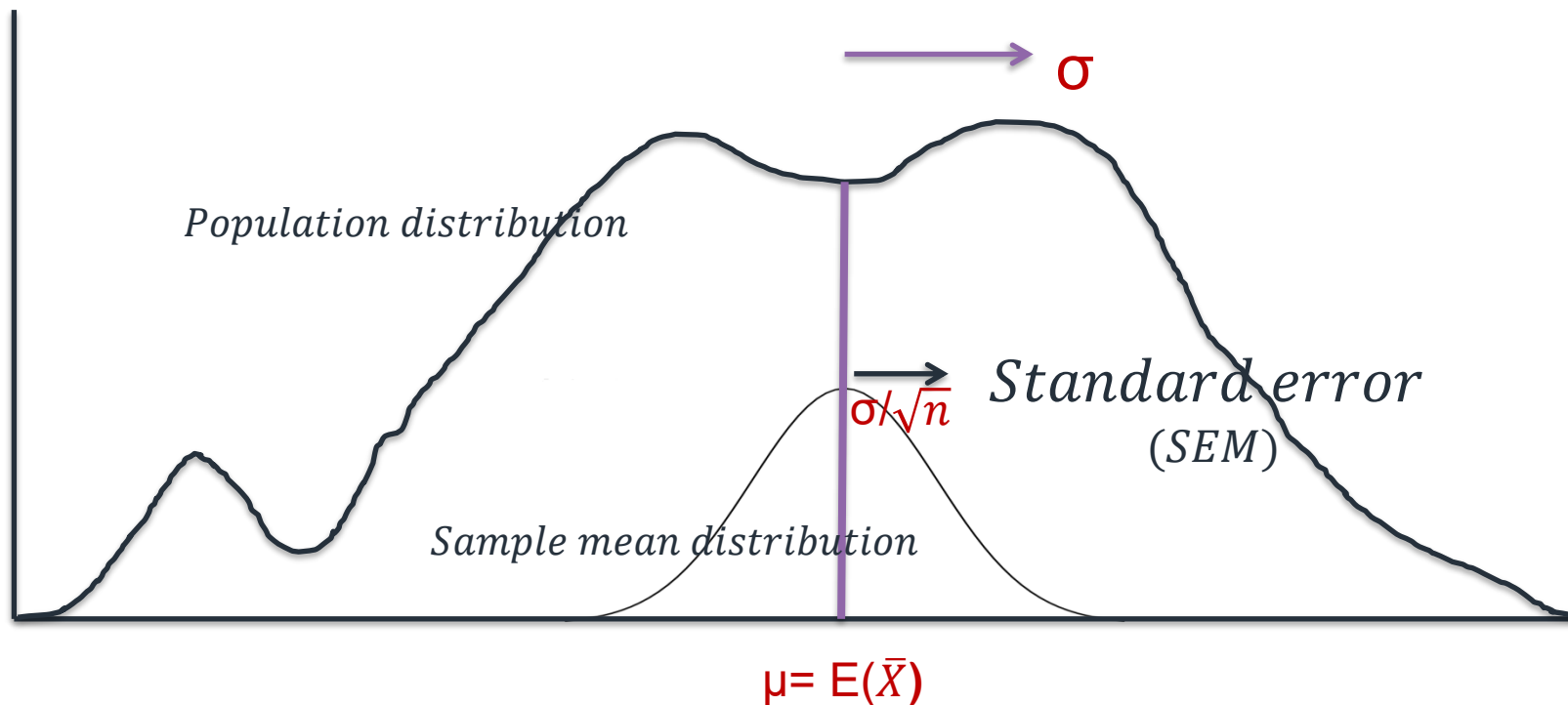
- The larger the sample size n is, the closer the distribution of the sample mean is to the normal

The Central Limit Theorem (CLT)



- In practice **how large** must n be, to be considered sufficiently large?
- It depends on how **different** the distribution of the sample is from the normal
- Rule of thumb:
 - If **n is 30 or more**
 - we use the result from the CLT to approximate the distribution of the sample mean with a normal distribution

The Central Limit Theorem



Sampling distribution of the sample proportion

- Consider now that the characteristic, X , of interest in a population is dichotomous, $X=1$ or $X=0$
- Assume that the **proportion** of observing the characteristic of interest in the population is π
- We could consider all possible random samples of size n from a population with proportion π and calculate the **sample proportion**

$$p = \sum_{i=1}^n x_i / n$$

for all the samples

Sampling distribution of the sample proportion

We would observe that:

- The mean of the calculated sample proportions is the proportion of the population

$$E(\mathbf{P}) = \pi$$

- The variance of the calculated sample means is NOT the variance of the population, but

$$\text{var}(\mathbf{P}) = [\pi(1 - \pi)] / n$$

- We call $\text{SEP} = \sqrt{\text{var}(\mathbf{P})} = \sqrt{[\pi(1 - \pi)]/n}$ the **standard error of the sample proportion**

Sampling distribution of the sample proportion

- From the CLT, when the sample size is large, the **distribution of the sample proportion** is **approximately normally distributed** with
 - mean π and variance $[\pi(1-\pi)] / n$
- How large should the sample size n be?
- Rule of thumb:
- We can use the CLT approximation when
$$n\pi \text{ and } n(1-\pi) \text{ are both greater than 5}$$

Point Estimation

Inference

- **Statistical inference** is the procedure by which we reach a conclusion about a population based on the information contained in a sample drawn from that population
- Warning:
 - Random **sample** drawn from **sampld population**
 - **Inference valid** for the sampled population
 - Sampled population **might be different** from the target population

Estimation

- Point estimates
 - Interval estimates
-
- A **point estimate** is calculated from the data of a sample. It is a **single value** used as the best guess of a corresponding population parameter
 - An **estimator** is the **procedure** or **rule** used to calculate an estimate

Example: the sample mean estimator

$$\bar{X} = \frac{\sum X_i}{n}$$

Good estimators

- There are several **criteria** for judging estimators
- Here we will use **unbiasedness**
- An estimator is an **unbiased estimator** of a population parameter if the mean of the statistic calculated from all possible samples is the population parameter, i.e., if the expected value of the estimator is the parameter

Examples: Sample mean: we have seen that $E(\bar{X}) = \mu$

Sample proportion: $E(\mathbf{P}) = \pi$

- Hence, $\bar{X}$ and $\mathbf{P}$ are **unbiased estimators** of the corresponding population parameters μ and π respectively

Good estimators

- What about the **sample variance**?

- The sample variance estimator calculated as

$$\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n}$$

systematically underestimates the population variance and hence it is NOT an unbiased estimator

- An **unbiased estimator** of the **population variance** σ^2 is

$$s^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n - 1}$$

Good estimators

Example: Consider the dataset

(1.69,1.7,1.72,1.72,1.73,1.75,1.76,1.8,1.81,1.83,1.83,1.84,1.89,1.9,1.93,1.54,1.58,1.64,1.69,1.73)

- The variance of the dataset calculated as

$$\sigma^2 = \frac{\sum_{i=1}^n (x_i - \mu)^2}{N} = 0.009934, \quad \text{where } \mu = 1.754$$

describes well the **spread of the dataset** but it is a **biased estimate** of the **variance of the population** from which it was drawn

- If the dataset is a sample, then an **unbiased estimate** of the **population variance** σ^2 is

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1} = 0.010456$$

which is larger

Which of these use s , NOT σ ?

- Using the weights of **some** of the parcels going through the post room to learn about the rest.
- Trying to estimate the spread of heights of **all** 1st year students from **your** height data.
- Describing the spread of heights **in this** class.



Interval Estimation

Estimation

- Point estimates
- Interval estimates
- An **interval estimate** is an **interval** that we can say with a **degree of confidence** that the parameter of interest is contained in that interval
- We refer to interval estimates as **confidence intervals** and to the bounds of the confidence interval as **confidence limits**

Confidence levels



- When calculating confidence intervals, you need to decide what **level of confidence** to use
- The most common confidence levels used by researchers are: 90%, **95%**, 98%, 99%
- **A 95% confidence level means that there is a 95% chance that the true population parameter lies within the confidence interval (CI); i.e. there is a 5% chance that the true parameter value does not lie in the confidence interval**
- The chance that the true parameter does not lie within the CI is called alpha (α)

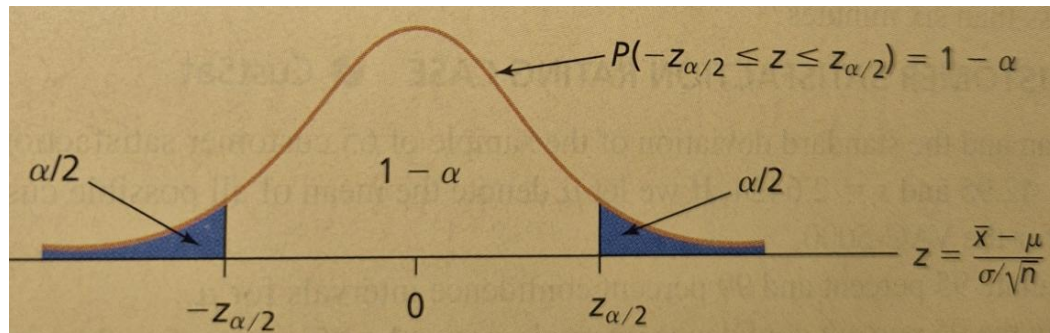
Large sample confidence interval for the population mean

- If the population follows a **normal distribution** with mean μ and variance σ^2 , then we know that the sampling distribution of the sample mean, $\bar{X}$, has a normal distribution with $E(\bar{X}) = \mu$ and $\text{var}(\bar{X}) = \sigma^2 / n$

- Hence, $z = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}}$ has a standard normal distribution

- And the probability that

$$-Z_{\alpha/2} \leq \frac{\bar{X} - \mu}{\sigma / \sqrt{n}} \leq Z_{\alpha/2} \quad \text{is } (1-\alpha)$$

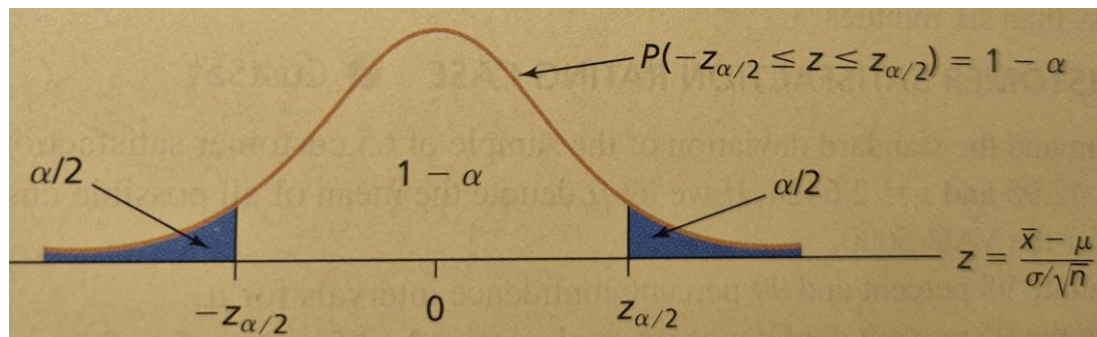


Large sample confidence interval for the population mean

- It follows that the probability that the population mean μ is between

$$\bar{x} - z_{\frac{\alpha}{2}} \times \frac{\sigma}{\sqrt{n}} \quad \text{and} \quad \bar{x} + z_{\frac{\alpha}{2}} \times \frac{\sigma}{\sqrt{n}}$$

is $(1-\alpha)$

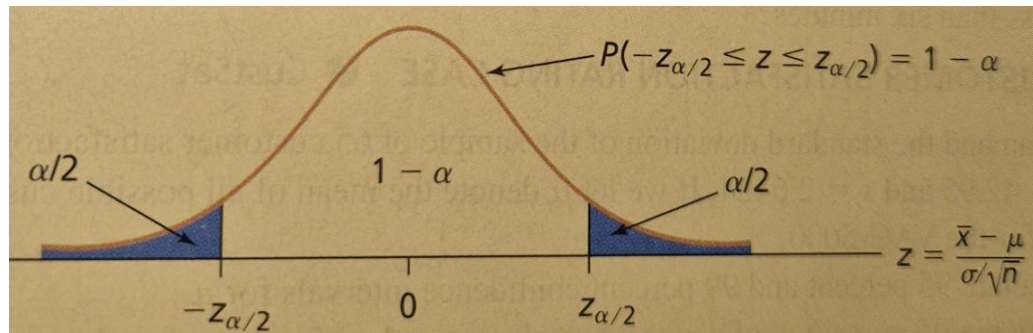


Large sample confidence interval for the population mean

- If the population does **not** follow a **normal distribution**
- We can use the Central limit Theorem (large sample) and say that
- Approximately, the probability that the population mean μ is between

$$\bar{x} - z_{\frac{\alpha}{2}} \times \frac{\sigma}{\sqrt{n}} \quad \text{and} \quad \bar{x} + z_{\frac{\alpha}{2}} \times \frac{\sigma}{\sqrt{n}}$$

is $(1-\alpha)$

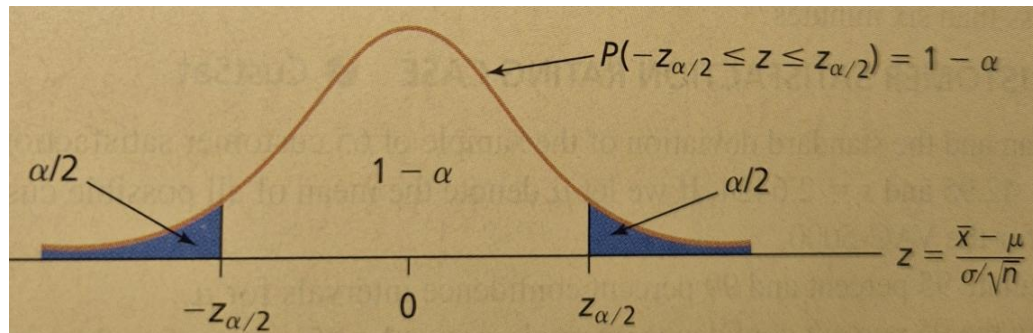


Large sample confidence interval for the population mean

- If the population σ is not known (usually isn't)
- Because the sample is large ($n \geq 30$)
- We can use the sample estimate s instead of the population σ in the $(1-\alpha)$ confidence limits:

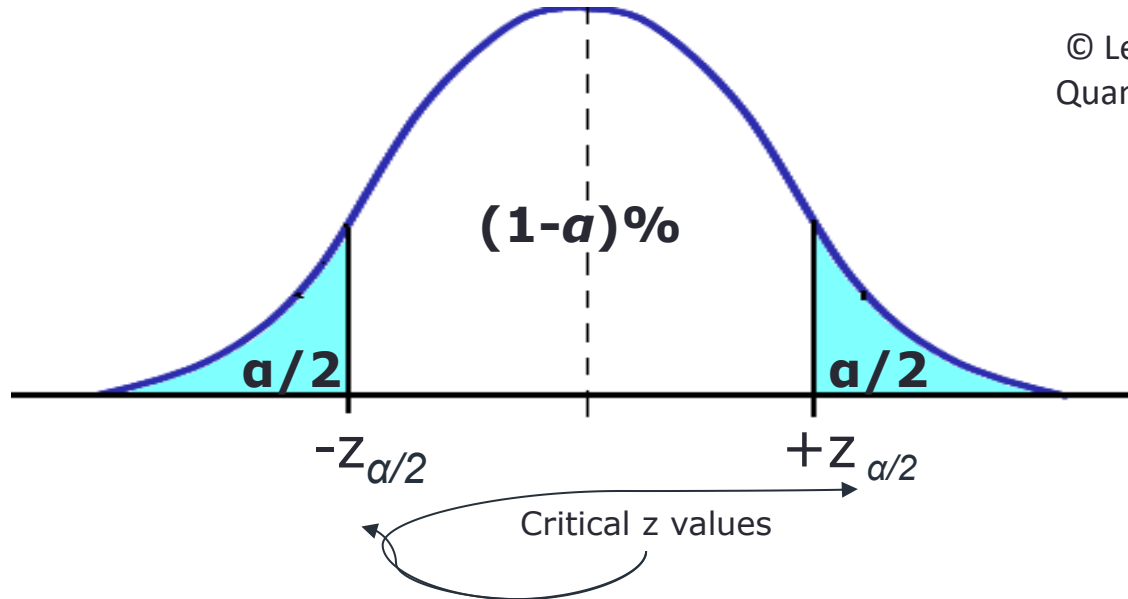
$$\bar{x} - z_{\frac{\alpha}{2}} \times \frac{s}{\sqrt{n}} \quad \text{and} \quad \bar{x} + z_{\frac{\alpha}{2}} \times \frac{s}{\sqrt{n}}$$

where $s = \sqrt{\frac{\sum(x - \bar{x})^2}{n-1}}$



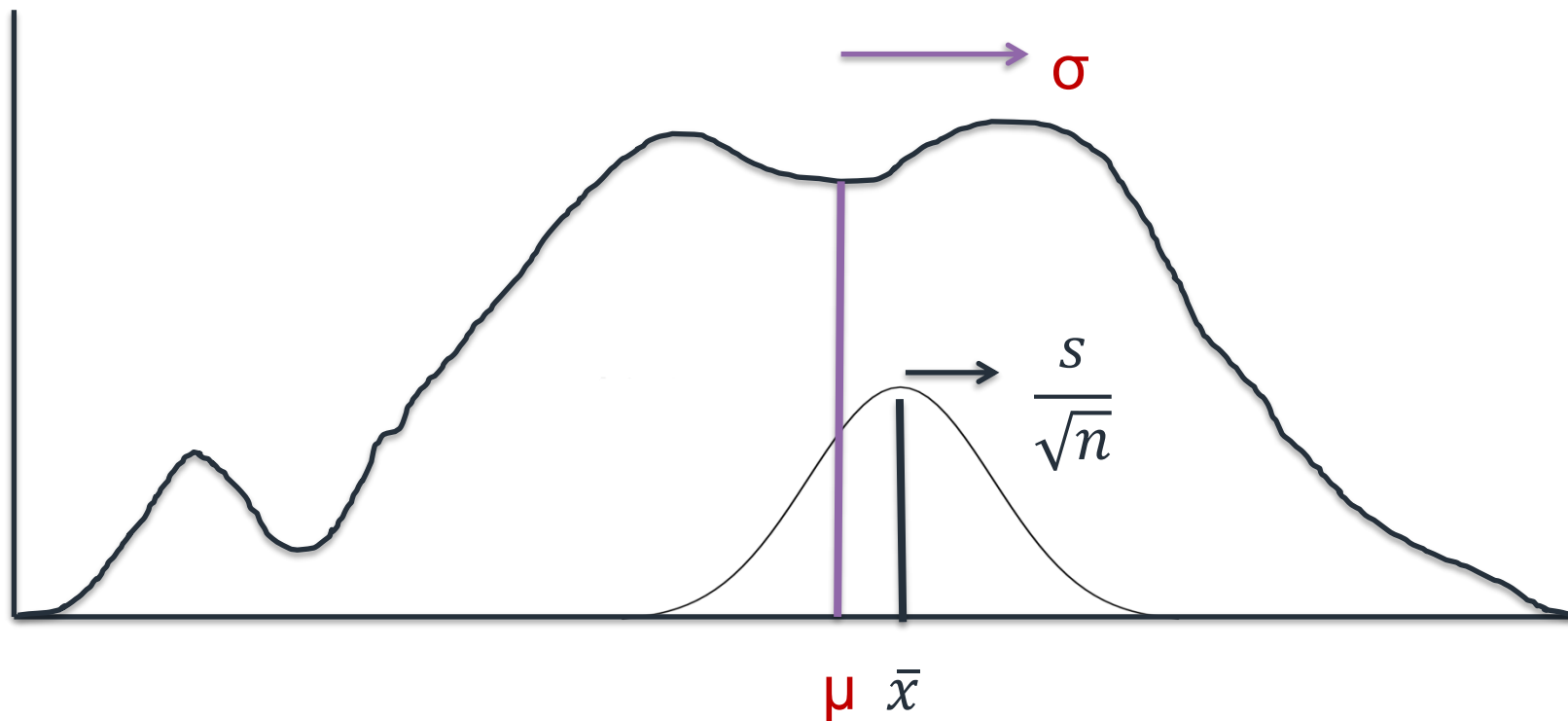
Confidence interval for a population mean for large samples

- $\alpha/2$ % is the area in one of the tail ends
- α % is the total area in both tails of the Normal distribution curve, outside the confidence interval area

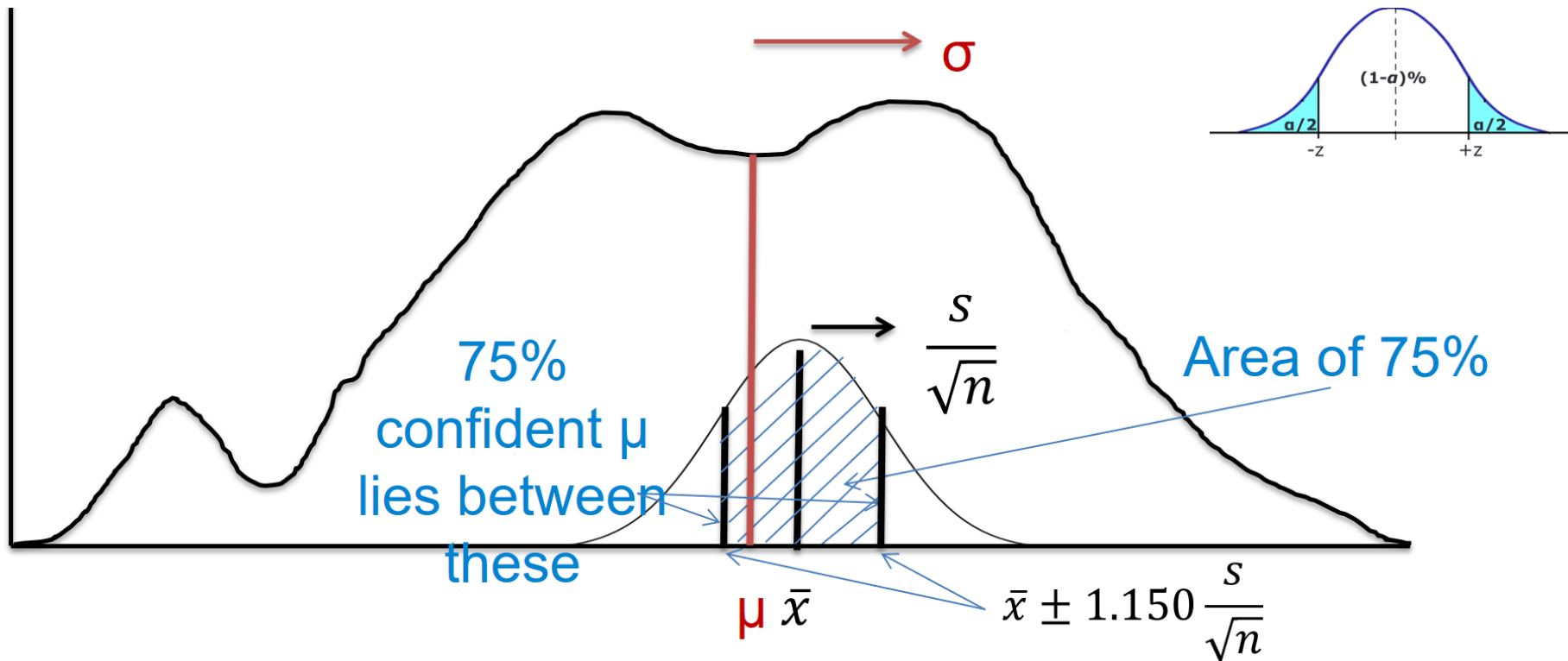


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In a picture



Confidence Interval example



Some key points from the last lecture

- Population & sample
- Sample should be chosen so that it is representative of the population
- We use a sample to infer something about a population

	Population parameter	Sample statistic
Mean	μ	$\bar{x}$
Standard deviation	σ	s
Proportion	π	p

Some key points from the last lecture

- The **sample mean** and **proportion** are the **best estimates** of the population parameters μ and π .
- These are called the ***Point estimates***.
- The **standard deviation** of a sample is a ***biased estimator***.
- To overcome this bias we use ***n-1*** and **not *n*** in the denominator of the formula for standard deviation.

Small sample confidence interval for the population mean

- What if the sample is **not large** enough ($n < 30$) ?
- As we usually don't know the variance of the population
- We can use the sampling distribution of $t = \frac{\bar{X} - \mu}{S / \sqrt{n}}$
- If the population is **normally distributed**, then (for any sample size n) this sample distribution is what is called a **t distribution**
- The confidence interval is $\bar{x} \pm t_{\frac{\alpha}{2}} \times \frac{s}{\sqrt{n}}$

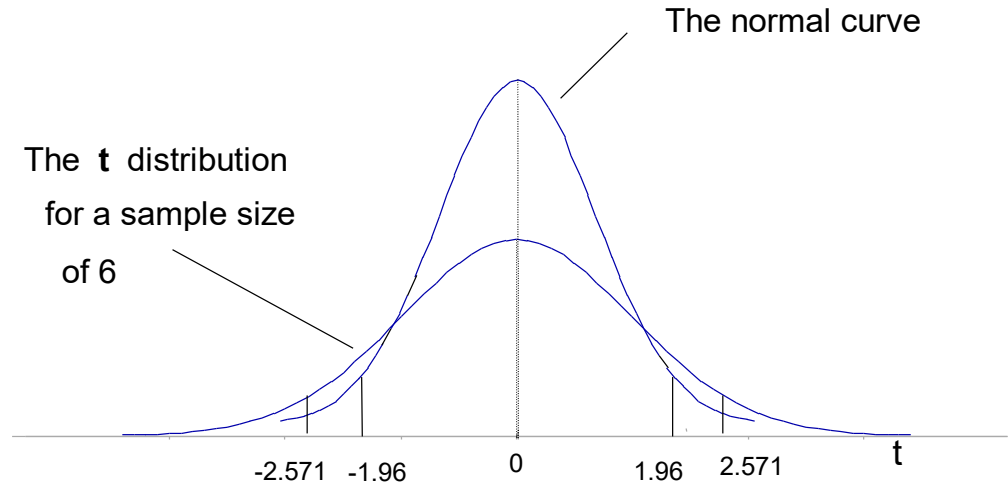
Obs: If the population is not normally distributed and $n < 30$ the possible methods are not covered here, e.g. bootstrap, or data transformation.

The t-distribution

- The spread of the t distribution depends on a parameter called the **number of degrees of freedom**, denoted ***df***
- the sampling distribution of

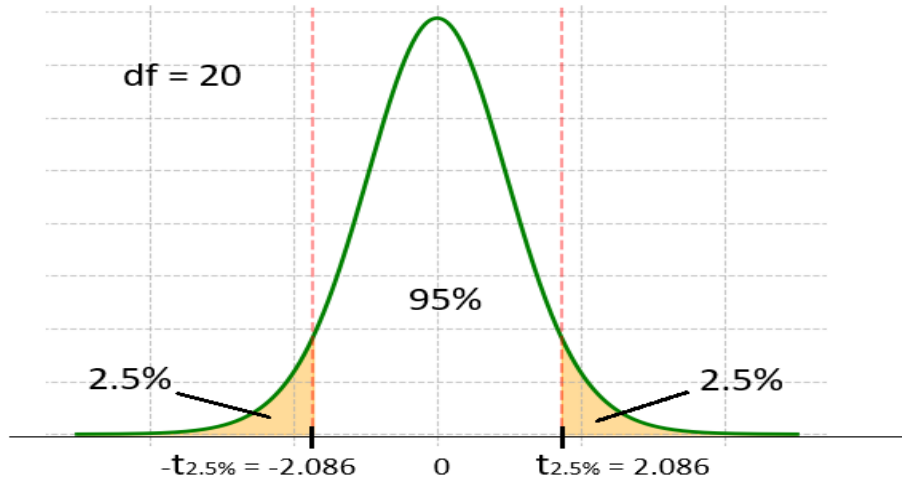
$$t = \frac{\bar{X} - \mu}{S / \sqrt{n}}$$

- is a **t distribution**
- with **n-1** degrees of freedom



The t-distribution

- Left column: degrees of freedom
- Top row: 'One tail' is the probability to the right of the value in the table



PERCENTAGE POINTS OF THE T-DISTRIBUTION

One Tail	5%	2.5%	1%	0.5%	0.1%	0.05%
Two Tails	10%	5%	2%	1%	0.2%	0.1%
1	6.314	12.706	31.821	63.657	318.309	636.619
2	2.920	4.303	6.965	9.925	22.327	31.599
3	2.353	3.182	4.541	5.841	10.215	12.924
4	2.132	2.776	3.747	4.604	7.173	8.610
5	2.015	2.571	3.365	4.032	5.893	6.869
6	1.943	2.447	3.143	3.707	5.208	5.959
7	1.895	2.365	2.998	3.499	4.785	5.408
8	1.860	2.306	2.896	3.355	4.501	5.041
9	1.833	2.262	2.821	3.250	4.297	4.781
10	1.812	2.228	2.764	3.169	4.144	4.587
11	1.796	2.201	2.718	3.106	4.025	4.437
12	1.782	2.179	2.681	3.055	3.930	4.318
13	1.771	2.160	2.650	3.012	3.852	4.221
14	1.761	2.145	2.624	2.977	3.787	4.140
15	1.753	2.131	2.602	2.947	3.733	4.073
16	1.746	2.120	2.583	2.921	3.686	4.015
17	1.740	2.110	2.567	2.898	3.646	3.965
18	1.734	2.101	2.552	2.878	3.610	3.922
19	1.729	2.093	2.539	2.861	3.579	3.883
20	1.725	2.086	2.528	2.845	3.552	3.850
25	1.708	2.060	2.485	2.787	3.450	3.725
30	1.697	2.042	2.457	2.750	3.385	3.646
40	1.684	2.021	2.423	2.704	3.307	3.551
50	1.676	2.009	2.403	2.678	3.261	3.496
60	1.671	2.000	2.390	2.660	3.232	3.460
∞	1.645	1.960	2.326	2.576	3.090	3.291

The t-distribution

- The t-distribution(s) are very much like the normal distribution, except that they have **thicker tails**, that is, they are **more spread out**. (Higher kurtosis, they are leptokurtic.)
- They are used for creating confidence intervals when the sample size is “**small**” or when you **don’t know** the population standard deviation σ .

- Summary: CI for μ

	Large Sample (Population Normal or not)	Small Sample (Population Normal)
Know σ	Normal (z)	Normal (z)
Only have s	Normal (z)	t

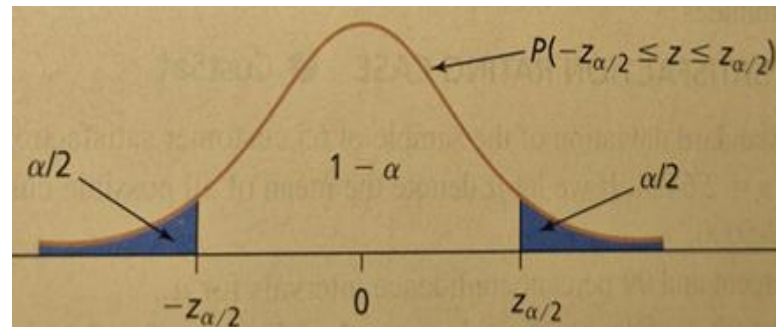
Confidence interval for the population proportion

- When np and $n(1-p)$ are both **greater than 5**, and n is **at most 5%** of the population size N , the approximate **(1- α) confidence interval** for the population proportion π is given by

$$p \pm z_{\frac{\alpha}{2}} \times \sqrt{\frac{p(1-p)}{n}}$$

- If np and $n(1-p)$ are greater than 5, but n is more than 5% of the population size N , include a finite population correction:

$$p \pm z_{\frac{\alpha}{2}} \times \sqrt{\frac{p(1-p)}{n} \cdot \frac{N-n}{N-1}}$$



In general

The format of a confidence interval is:

Point Estimate $\pm$ **Confidence Coefficient** x Standard Error

(PE $\pm$ CC x SE)

(We call the CC x SE part the **Margin of Error**)

That is, one of:

$$\bar{x} \pm CC \times \frac{s}{\sqrt{n}}$$

$$p \pm CC \times \sqrt{\frac{p(1-p)}{n}}$$

An example

- A chemical process yields, over a sample of runs, products with masses:
2.3mg, 3.9mg, 1.9mg, and 2.7mg.
- Calculate the 95% confidence interval for the true **mean** mass of the yield.

An example

- So, first we need to calculate is the mean and (inferential) standard deviation of the sample: 2.3mg, 3.9mg, 1.9mg, and 2.7mg.

- These end up being: (Check you can replicate these!)

- $\bar{x} = 2.7\text{mg}$ and

- $s = 0.864\text{mg}$ respectively.

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$$

Diagram illustrating the formula for the confidence interval:

$$\bar{x} \pm \left[t_{\frac{\alpha}{2}, n-1} \times \frac{s}{\sqrt{n}} \right]$$

Labels and arrows:

- $\bar{x}$ is labeled "Sample mean" (green arrow).
- $t_{\frac{\alpha}{2}, n-1}$ is labeled "CC=t value for $\frac{\alpha}{2}$ & df = n-1" (red arrow).
- $\frac{s}{\sqrt{n}}$ is labeled "SE" (blue arrow).

An example

- We also need to find the confidence coefficient.
- We're in a 2-tail situation (confidence interval), we want 95% confidence, and $n=4$, so we look at row 3.
- That leads us, from the table, to 3.182.
- (Again, check you can replicate this.)

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60	1.671	2.000	2.390	2.660	3.232	3.460
∞	1.645	1.960	2.326	2.576	3.090	3.291

An example

We then have to do the math:

$$\begin{aligned} & \bar{x} \pm CC \frac{s}{\sqrt{n}} \\ &= 2.7\text{mg} \pm 3.182 \frac{0.864\text{mg}}{\sqrt{4}} \\ &= 2.7\text{mg} \pm 3.182 \times 0.432\text{mg} \\ &= 2.7\text{mg} \pm 1.375\text{mg} \end{aligned}$$

$$CI = [1.33\text{mg} , 4.07\text{mg}]$$

Which is our 95% confidence interval for the true mean.

Determining Sample Sizes

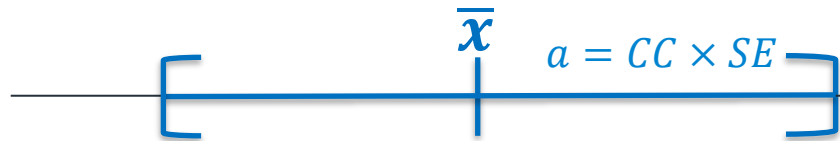
Determining Sample Sizes

- We can also run all this backwards. That is, rather than knowing $\bar{x}$, s , n and whatever confidence level we want and then finding CC (confidence coefficient) and SE...
- We can pick the precision (width) of the confidence interval we want (that is, $CC \times SE$, the margin of error) as well as the confidence level we want and find what n should be.
- If the sample size n is small then the CI is very wide (low precision, high margin of error)
- If the sample size n is very large it can be unnecessarily expensive, time consuming

Example: “I want to know the public’s opinion of this topic with high precision (low margin of error). How many people should I ask?”

Determine sample size for estimating means

- What is the precision desired? How narrow do we want the CI?
- How much confidence we want for the interval? Confidence coefficient, CC?
- How much variability is there in the population, σ ?
- The CI is $\bar{x} \pm CC \times \frac{s}{\sqrt{n}}$ and $CC \times \frac{s}{\sqrt{n}}$ is twice the width of the CI
- Set the precision (margin of error) $a = CC \times SE = CC \times s/\sqrt{n}$, then
$$n = \left(\frac{CC \times s}{a} \right)^2$$
- The CI level of confidence determines the CC
- We need a pilot study to estimate s



Example: sample size for mean

- Jo Cabot is looking to move into insuring houses. A quick analysis of some relevant house prices says that the **mean price is £213,748 with a standard deviation of £61,093 (s)**.
- How many houses should Jo assess to know the mean value to **within £20,000** (margin of error)?

Example: sample size for mean

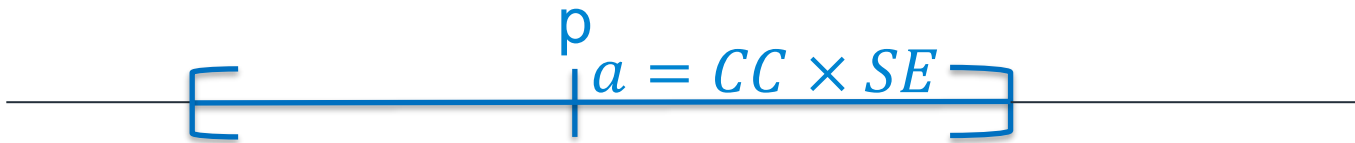
We have our estimated standard deviation and desired margin of error. We default to 95% confidence level ($z_{0.025} = t_{0.025, \infty} = 1.96$)

$$n = \left(\frac{CC \times s}{a} \right)^2 = \left(\frac{1.960 \times 61,093}{20,000} \right)^2 = 35.8$$

(We round up to 36.)

Although it might be worth trying 99% if Jo is very risk averse.
For 99% confidence, we use 2.576 rather than 1.960 and find $n=62$.

Sample size for estimating proportions

We set 

$a = \text{Margin of Error} = CC \times SE$

so

$$a = CC \times \sqrt{\frac{p(1-p)}{n}} \Rightarrow n = \frac{CC^2 p(1-p)}{a^2}$$

(If we don't have an estimate for p , use 0.5.)

(Two-tails, infinity row for CC .)

Example: sample size for proportion



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“I want to know the public’s opinion of this topic within 5% (margin of error) with 95% confidence level (Confidence Coefficient). How many people should I ask?”

$\alpha = 0.05$ (5%), $CC = 1.960$ (2-tails, 5%, ∞ row)

$$n = \frac{CC^2 p(1-p)}{\alpha^2} = \frac{1.960^2 0.5(1-0.5)}{0.05^2} = 384.16.$$

384 people would give slightly less than 95% confidence or an accuracy slightly worse than 5%, so we say 385.

Exercise:

Suppose that you want to estimate the **average age** of all Boeing 727s now in active domestic US service. You want to be 95% confident (CC) and you want your investigation to be within 2 years (margin of error) of the actual figure (the average age). The standard deviation of the population is 6.25.

How large a sample should you take?

Exercise:

We have our estimated standard deviation and desired margin of error. We default to 95%

$$n = \left(\frac{CCs}{a} \right)^2 = \left(\frac{1.960 \times 6.25}{2} \right)^2 = 37.515625$$

(We round up to 38.)

Summary of the topic

We looked at:

- Sampling
- Sampling distributions
- The Central Limit Theorem
- Point estimation
- Confidence intervals

(Oakshott, Ch3 & Ch8)

Next topic

Testing statistical hypotheses

Reading: Chapter 9 from the core-text